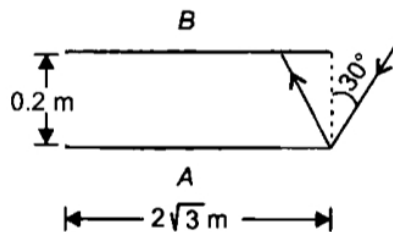




LIGHT -05

Class 10 - Science

1. The formula which gives the relationship between the object distance, image distance and focal length of the spherical mirror is: **[1]**
a) $\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$
b) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
c) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
d) $\frac{1}{v} = \frac{1}{u} + \frac{1}{f}$
2. The real image formed by a concave mirror is larger than the object when the object is: **[1]**
a) between focus and centre of curvature
b) at a distance less than the focal length
c) at a distance greater than radius of curvature
d) at a distance equal to radius of curvature
3. Your school laboratory has one large window. To find the focal length of a concave mirror using one of the walls as the screen, the experiment may be performed: **[1]**
a) on the wall adjacent to the window
b) on the same walls as the window
c) only on the table as per the laboratory arrangement
d) near the wall opposite to the window
4. If the focal length of a spherical mirror is 12.5 less cm, its radius of curvature will be: **[1]**
a) 35 cm
b) 20 cm
c) 15 cm
d) 25 cm
5. Two plane mirrors A and B are aligned parallel to each other as shown in the figure. A light ray is incident at an angle 30° at a point just inside one end of A. The plane of incidence coincides with the plane of the figure. The maximum number of times the ray undergoes reflections (including the first one), before it emerges out, is **[1]**



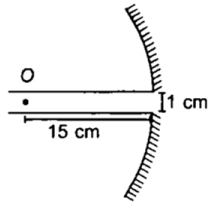
- a) 28 b) 30
- c) 32 d) 34
6. Under which of the following conditions a concave mirror can form an image larger than the actual object? [1]
- a) When object is kept at a distance greater than its radius of curvature b) When object is placed between the focus and centre of curvature
- c) When the object is kept at a distance equal to its radius of curvature d) When object is kept at a distance less than its radius of curvature

to its radius of curvature

its focal length

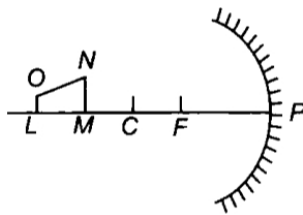
7. What is not a characteristic of a rearview mirror in a car? [1]
- a) They give a virtual image b) Convex in nature
- c) Concave in nature d) They have wider field of coverage

8. A mirror is broken into two parts and these parts are separated by a distance of 1 cm as shown in figure. The focal length of the mirror is 10 cm. The height of the images (in cm with respect to their optic axis), formed by the two parts of mirror when the object is midway between the two principal axes are [1]

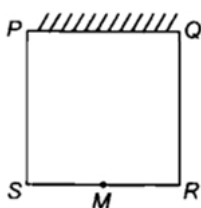


- a) 2,2 b) 0.5,1
- c) 1,2 d) 1,1
9. Magnification produced by a rear view mirror fitted in vehicles [1]
- a) is equal to one b) can be more than or less than one depending upon the position of the object in front of it
- c) is less than one d) is more than one

10. An object LMNO is placed in front of a concave mirror beyond the centre of curvature C as shown in figure. Which of the following is correct regarding the shape of the image? [1]



- a) $|m_{LO}| < |m_{MN}|$ b) $|m_{LO}| < 1$ and $|m_{MN}| < 1$
- c) Both $|m_{LO}| < 1$ and $|m_{MN}| < 1$ and $|m_{LO}| < |m_{MN}|$ d) $|m_{LO}| > |m_{MN}|$
11. PQRS shows the walls of room each measuring 8 m, such that wall PQ has a full length plane mirror. A camera is placed at point M, the mid point of wall SR. In order to take clear image of the point R, the camera should be focused at. [1]



- a) More than 16 metres b) 16 m
- c) 8 m d) 4 m
12. To obtain a magnification of +2 with a concave mirror of radius of curvature 60 cm the object distance must be [1]
- a) -90 cm b) -45 cm

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- c) 10 m d) 20 m

31. When an object is kept within the focus of a concave mirror, an enlarged image is formed behind the mirror. This image is: **[1]**

a) virtual and erect b) inverted

c) real d) virtual and inverted

32. Find the focal length of a convex mirror of radius of curvature 1 m. **[1]**

a) 0.25 m b) None of these

c) 0.5 m d) 1 m

33. In which of the following, the image of an object placed at infinity will be highly diminished and point sized? **[1]**

a) Concave mirror only b) Convex lens only

c) Concave mirror, convex mirror, concave lens and convex lens d) Convex mirror only

34. A piece of red cloth when suitably illuminated may look black, but a piece of black cloth will never appear red. This phenomenon occurs because: **[1]**

a) Red cloth reflects all colours b) Black cloth reflects all colours

c) Black cloth reflects only black light d) Black absorbs all the colours

35. A spherical mirror and a thin spherical lens have each a focal length of -15 cm. The mirror and the lens are likely to be **[1]**

a) The mirror is convex, but the lens is concave. b) Both convex

c) Both concave d) The mirror is concave and the lens is convex.

36. If R is the radius of curvature of a spherical mirror and f is its focal length then: **[1]**

a) $R = f$ b) $R = \frac{f}{2}$

c) $R = 3f$ d) $R = 2f$

37. The image formed by a concave mirror is real, inverted, and highly diminished (much smaller than the object). The object must be: **[1]**

a) at infinity b) at the centre of curvature

c) between pole and focus d) at focus

38. To determine the focal length of a concave mirror, a student focuses a distant object using the concave mirror. The best object can be: **[1]**

a) a distant tree b) All of these

c) classroom window d) sun

39. The image of an extended object placed in front of a concave mirror is formed at a distance of 40 cm from the object. If the image is 3 times bigger than the object, the magnitude of focal length of the mirror is **[1]**

a) 5 cm b) 10 cm

c) 20 cm

d) 15 cm

40. In the concave reflector of a torch, the bulb is placed: [1]

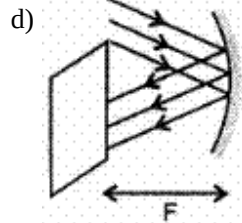
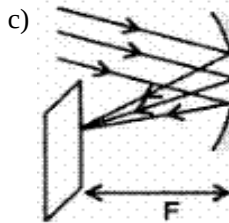
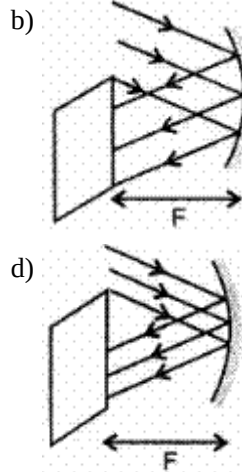
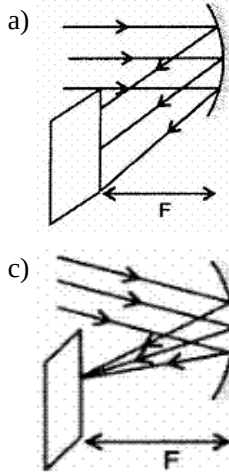
a) between focus and centre of curvature of reflector

b) at the focus of reflector

c) between the pole and focus of reflector

d) at the centre of curvature of reflector

41. Which of the following pictures depict the correct image formation [1]



42. The focal length of a small concave mirror is 2.5 cm. In order to use this concave mirror as a dentist's mirror, the distance of tooth from the mirror should be: [1]

a) 4.5cm

b) 1.5cm

c) 3.5cm

d) 2.5cm

43. The angle between an incident ray and the plane mirror is 30° . The total angle between the incident ray and reflected ray will be: [1]

a) 120°

b) 90°

c) 60°

d) 30°

44. A full length image of a distant tall building can definitely be seen by using [1]

a) both concave as well as plane mirror

b) a plane mirror

c) a concave mirror

d) a convex mirror

45. Which of the following lenses would you prefer to use while reading small letters found in a dictionary? [1]

a) A convex lens of focal length 50 cm

b) A convex lens of focal length 5 cm

c) A concave lens of focal length 50 cm

d) A concave lens of focal length 5 cm

46. The angle of incidence for a ray of light passing through the centre of curvature of a concave mirror is: [1]

a) 45 degree

b) 90 degree

c) 180 degree

d) 0 degree

47. A man sits in an optician's chair, looking into a plane mirror which is 2 m away from him and views the image of a chart that faces the mirror and is 50 cm behind his head. How far away from his eyes does the chart appear to be? [1]

a) 1.5 cm

b) 2.5 cm

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He will observe that the screen is on the same side as the object

- a) in both the cases
- b) in case (ii) but not in case (i)
- c) in case (i) but not in case (ii)
- d) in neither of the two cases

58. A man stands 10 m in front of a large plane mirror. How far must he walk before he is 5 m away from his image? [1]

- a) 8.5 m
- b) 8.0 m
- c) 7.0 m
- d) 7.5 m

59. A student obtains a blurred image of an object on a screen by using a concave mirror. In order to obtain a sharp image on the screen, he will have to shift the mirror. [1]

- a) towards the screen
- b) to a position very far away from the screen
- c) depending upon the position of the object
- d) away from the screen

60. An object is placed 20 cm in front of a plane mirror. The mirror is moved 2 cm towards the object. The distance between the positions of the original and final images seen in the mirror is: [1]

- a) 2 cm
- b) 10 cm
- c) 4 cm
- d) 22 cm

61. If a pencil beam is allowed to fall along the principal axis of a concave mirror, the ray will [1]

- a) emerge out along the principal axis
- b) deviate by 60°
- c) retrace its path along the principal axis
- d) deviate by 30°